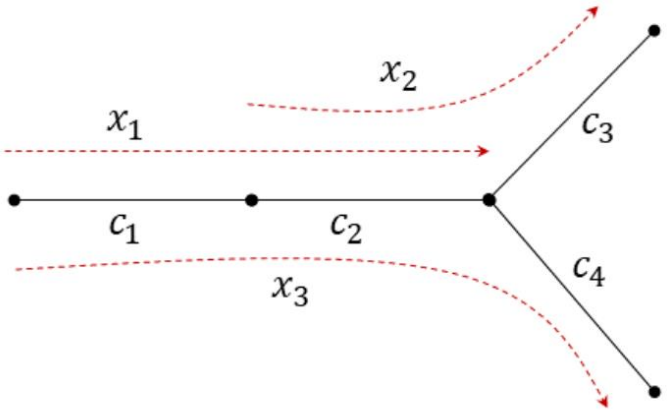
Lab DUAL DECOMPOSITION



Let's $c_1 = 10, c_2 = 6, c_3 = 10, c_4 = 8$ Mbps. Using alpha-fair utility family

$U(x) = (x^{1-\alpha})/(1-\alpha)$ if $\alpha > 0, \alpha \neq 1$, $U(x) = log(x)$ if $\alpha = 1$.

Write a program running the dual decomposition algorithm to solve for the optimal x_1, x_2, x_3 in the above network.

Show the results with different α values: 1, 2, 4. Observe the rate allocation at the bottleneck link (link 2).

- END -